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Commentary

A multidisciplinary team and multiagency approach for AI implementation: A commentary for medical imaging and radiotherapy key stakeholders

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Introduction

Artificial Intelligence (AI) algorithms have been recently deployed in different healthcare settings. These tools have shown promise in reducing professionals' administrative workload, handle electronic health records, aid drug discovery, improve diagnostic services, and analyse complex data [1-3]. Medical Imaging and Radiotherapy (MIRT) are at the forefront of this digital transformation, as can be seen from the concurrent increase in MIRT-related AI products [4-6]. In MIRT in particular, AI has shown potential to optimise image acquisition and post-processing, redefine clinical workflows, improve diagnostic accuracy, facilitate automated organ segmentation, image registration and planning in radiotherapy, and personalise patient care [7-9]. These recent advancements could translate into improved patient outcomes, personalised pathways and treatment plans, and, therefore, deliver precision into health-

care [10,11]. At the same time, concerns about potential risks and safety of AI-enabled software and hardware are raised by both clinical practitioners and patients, which need to be addressed within the design and implementation of AI and balanced against the undoubted benefits [12].

Being at the front of technological advancements comes at a cost. As AI applications emerge, are tested and rigorously evaluated in clinical places, MIRT professionals are responsible for the fine-tuning, co-ordination, and governance of the clinical implementation of AI. While various professions, agencies and stakeholders in related disciplines work to understand what impact and change AI implementation might have on clinical workflows, their future roles and careers, and to address local challenges in knowledge gaps, training, and the wider workforce, interprofessional coordination is of paramount importance [13].

The art (and science) of change in healthcare

Implementing a novel technology, like AI, requires adequately considering and addressing the impact on workflows, communication pathways, and professional roles and interactions. Failure to do so can both undermine the potential benefit to care but also introduce stress and lead to implementation delays or failures. It is thus important to afford explicit consideration to the process and science of implementation when introducing change [14]. The importance of doing so in a collaborative way has been demonstrated in the introduction of other disruptive technologies in MIRT in recent years, including Image-Guided Radiation Therapy (IGRT) [15,16], and picture and archiving communication systems (PACS) [17].

The different professionals in the new multidisciplinary, interprofessional teams will also need to communicate in a common language, in this case the language of AI terminologies. It is, therefore, imperative for them to be educated on the requirements, potential and risks of these new technologies [13, 18-23]. It is, furthermore, vital to agree upon a common ethics and governance framework [13,24-26].

In addition, the mediation and collaboration of professional bodies in this ecosystem is paramount for governance and safe deployment. Both uni-disciplinary professional bodies, like the Society and College of Radiographers [27], the Royal College of Radiologists [28], the Institute of Physics and Engineering in Medicine [29] and multi-professional ones, for example, the British Institute of Radiology (BIR) in the UK [30] and the European Society of Medical Imaging Informatics (EuSoMII) in Europe [31], have a role in this smooth transition. These bodies, while serving their membership, can empower and facilitate multidisciplinary team collaboration in AI research projects, enable the inclusion of public contributors, align policy and practice and provide some related training or signpost to relevant educational providers. In Canada, professional organizations in radiation medicine representing radiation oncologists, medical physicists, and medical radiation technologists (radiographers), have joined forces in a formal collaboration known as the Canadian AI and Data in Radiotherapy Alliance (CADRA) [32] which aims to consider system level data standardisation and coordination of AI adoption. Similar schemes exist in other countries.

The importance of interprofessional collaboration within multidisciplinary teams for AI integration

Interprofessional collaboration is essential to manage change and to provide well-functioning, high quality healthcare [33] and can be defined as “...when two or more professionals work together collaboratively to improve patient service delivery” [34]. The greatest value of interprofessional collaboration lies in the potential of offering multiple expert perspectives on ‘certain topics, such as patient care issues’ to enhance patient care and optimise healthcare delivery. Hence the different professionals can ‘learn from with, from and about each other to improve collaboration and quality of

care [35]. Interprofessional collaboration is achieved through multi-disciplinary teams. The importance of multidisciplinary work is further highlighted by recent studies showing that early involvement of key stakeholders is central to the successful and safe AI implementation in healthcare [36].

To ensure a smooth transition into the new digital era and to successfully and safely implement AI in clinical practice, it is imperative to empower and support multidisciplinary teams and foster interprofessional collaboration [21,37,38]. Multidisciplinary representation should be employed throughout the AI products’ lifecycle: when designing AI-enabled tools for healthcare services [39], during drug development [40], and more importantly for AI adoption at both local and systems level [41,42].

Representation is a key aspect of interprofessional collaboration. Multidisciplinary teams should be truly diverse in gender, age, ethnicity, ability, neurodiversity. Diverse teams have proven valuable enablers in healthcare transformation projects and these teams are more likely to recognise and address biases in AI tools, improve AI algorithms, and ensure fairness and trustworthiness of healthcare provision for marginalised population [43]. Diversity, equity, and inclusion should be drivers to ensure an equitable and diverse AI-enabled future in MIRT that works for every patient and professional [44-46]. It has been proposed that multidisciplinary teams involved in AI implementation should at a minimum include knowledge experts, decision-makers, and users of AI tools [47]. It is also important that interprofessional collaborations integrate the technical, clinical, and operational aspects of AI implementation early in the process [48-50].

The evolving ecosystems in medical imaging and radiotherapy

The native clinical ecosystems, that were responsible from healthcare in the recent past, will have to change and expand to accommodate the new requirements of digital transformation. The AI ecosystems in MIRT would require not only medical doctors (radiologists, nuclear medicine consultants, oncologists, and referring consultants), nurses, allied health professionals (diagnostic and therapeutic radiographers, nuclear medicine technologists, sonographers), clinical scientists, medical physicists, but will also now need to more actively integrate new professions, like computer scientists, informaticians, implementation experts, behavioural theorists, and emerging professionals, that cut through traditional disciplines, like the technical physicians in the Netherlands [51-54]. All these professionals, as experts by training, will need to more actively engage and work with public contributors, who are experts by experience, such as the patients, their families/carers and the public. These are, after all, at the centre of healthcare delivery and their input is vital in designing new tools and services (Fig. 1).

It is challenging to predict which professionals will be responsible for which tasks as we transition into a new era of digital transformation. According to the archetypes developed in work commissioned by the National Health Service

(NHS) England, all healthcare professionals could be classified into five categories based on their role in AI technologies; as shapers, drivers, creators, embedders, or users [55]. Also, important to note the new role of “Digital Transformation Lead”, a person or, more likely, a team who act as change managers for all the technical, clinical and operational changes needed for successful implementation. When it comes to AI specifically, these roles could be also found under the name “AI champion” or “AI ambassador” [56]. Despite the distinct archetype definition, the scope is open as to which profession might evolve into what role and this will most certainly vary in different geographical regions and under different regulatory and governance frameworks. Although it is expected that some professions would more naturally gravitate towards specific archetypes, which are more closely aligned to their training and current clinical roles and skill-mix, the distribution will also vastly depend on an individual’s initiative, training and engagement with new technologies and on a robust team culture during AI integration. Soon it might not be uncommon that different professionals will work under the same archetype classification; this only underpins the importance of working together now and through this transformation, to make the transition as smooth as possible for the benefit of our patients.

The AI ecosystem should be interconnected, and a proposed relationship between its key stakeholders can be shown below

(Fig. 1). Of course, it should be mentioned that not all initiatives require everyone’s engagement at any one time.

Patients and the public at the heart of AI integration

Since the deployment of AI is centred on patient experience and outcomes, it is vital to ensure that patients, their caregivers, and the public are equal partners in AI improvement projects and that their involvement is authentic and occurs as early as possible. This will ensure services are not just accessible and patient-friendly. Early involvement of public contributors is a key asset that can save cost and time for any AI innovation and implementation project. Patient and public involvement (PPI) should be prioritised in all forms (whether advisory or co-production) when developing healthcare AI algorithms, as patients can act as experts with lived experiences and co-designers [57,58]. Recent research has shown that patients generally have great expectations from AI, as it can improve the care they receive and enhance their overall experience. Patients have expressed concerns and reservations in relation to AI, such as potential loss of autonomy and data management, that must be addressed by the multidisciplinary teams [59-62]. PPI should not be tokenistic, but rather evolve into an organic collaboration between patients, clinical practitioners and researchers [63]. It should also be governed by specific, commonly agreed, terms of reference, to ensure a framework

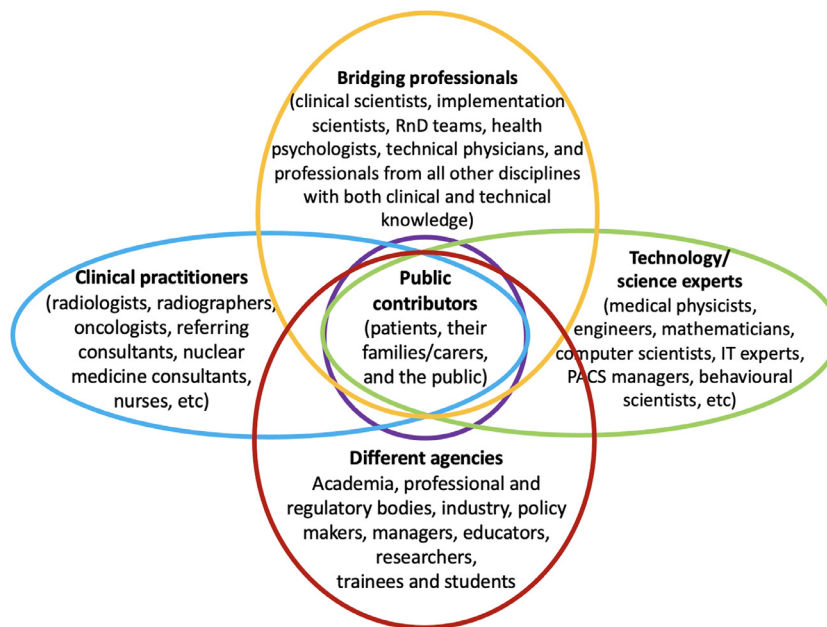


Fig. 1. An attempt to map out the different actors of the AI ecosystem in medical imaging and radiotherapy.

What ultimately connects them is their use of knowledge, skills and competencies to benefit public contributors (middle circle). This is not an exhaustive list of stakeholders, but one that attempts to show the complexity of a dynamic ecosystem in one healthcare domain (MIRT), where AI is currently actively deployed. Both clinical practitioners and technology/science experts could become bridging professionals with customised training, that extends beyond current educational schemes, norms and curricula. So, in this sense, this figure is not static. Bridging professionals with an appropriate skill mix will be ideally suited to act as AI ambassadors/AI champions, to facilitate and coordinate clinical implementation. Professionals like radiographers, who have both a clinical and technical element in their training, could be ideal candidates for these roles, but require robust knowledge of AI principles to be able to realise their full potential. Technical physicians are a new discipline piloted in the Netherlands, trying to bridge the gap between medical practice and digital technology. Radiographers in this graph represent a wide range of professionals working in medical imaging and radiotherapy, nuclear medicine and sonography. Other agencies beyond the clinical setting are important actors, to facilitate change management. Public contributors are a central part of this ecosystem.

of practice, where patients feel valued and their contributions are given enough time in team discussions and taken into account [64]. Equally, proportionate compensation of their time and effort [65], and flexible working patterns will ensure patients as actively engage co-researchers. This will enable the team to make decisions that reflect patients' priorities and avoid assumptions when developing, evaluating and implementing AI tools in healthcare [66]. The optimal method to work with practitioners and public contributors is co-production, "...an approach in which researchers, practitioners and the public work together, sharing power and responsibility from the start to the end of the project, including the generation of knowledge" [67]. Finally, patient groups that act as co-researchers in AI-related and other research projects need to also be diverse in gender, ability, neurodiversity, experience, age, digital competency, to ensure opinions are inclusive and represent the wider patient population and not a very specific subgroup. Different countries have different schemes for compensation of public contributors that act as co-researchers and different governance frameworks. In the UK, the National Institute for Health and Care Research (NIHR) INVOLVE guidelines are followed [68].

How can industry contribute: movers and shakers of AI

Change management teams should also include industry representatives. The enthusiasm, and know-how related to the diffusion of innovation, resources and agility industry bring could be useful enablers to drive clinical translation for AI and other innovations. They could support healthcare organisations overcome key challenges associated with the complexity of software and hardware, such as those arising from lack of interoperability or, often, lack of transparency of AI-enabled tools in MIRT equipment [69]. Healthcare industry are currently driving the fast pace that AI-enabled MIRT has assumed. Around 80% of the AI applications to the Food and Drug Administration (FDA) were from radiology companies at the end of 2023 [6]. There is great commercial interest in AI by the established MIRT manufacturers, but also by the new AI start-ups and small and medium-sized enterprises (SMEs).

However, sole reliance on private companies to provide AI products for use cases that may be publicly beneficial, but not commercially profitable, would, clearly, not be prudent. Industry-Academic-Clinical partnerships could converge efforts to ensure AI is properly governed, thoroughly tested, evidence-based and expertly deployed for a meaningful purpose, to solve real clinical problems. Industry can play an important role in maximizing transparency and explainability in the use of AI, for example, by providing clarity regarding the use cases of AI solutions, the population in which the algorithms were trained on, the thresholds and limits of intended use, and help visualise the impact on clinical workflows, patient diagnosis and treatment. Finally, industry could have a role in the monitoring of applications after their implementation.

What does involving the whole ecosystem look like?

With the overarching goal of successfully implementing AI in MIRT, while improving patient outcomes and experience to effectively involve the whole MIRT ecosystem, the approach of the wider AI ecosystem could be:

- ✓ **Multidisciplinary:** requires the team to bring together professionals from different fields of specialisation [70].
- ✓ **Interprofessional:** involves different professions aiming to collaborate for a common goal.
- ✓ **Transdisciplinary:** involves individuals with cross-disciplinary expertise to overcome the boundaries of various disciplines. It is beneficial because it aims to integrate all perspectives [71].
- ✓ **Cross-disciplinary:** engages in practices related with subjects outside the scope of a discipline, but without any integration from other disciplines. It has proven particularly beneficial for knowledge exchange in areas where medical science and social sciences meet [72].
- ✓ **Interdisciplinary:** requires professionals to synthesise approaches taken from different disciplines to integrate knowledge [73].
- ✓ **Multiagency** approaches should also be employed for AI implementation; these require professionals from several organisations and public to work together. For instance, academia, industry, professional bodies, public contributors, managers, trainees are proven to be valuable "agencies: to support the work of clinical practitioners.

To summarise, AI implementation teams should be comprised of professionals from different fields; work beyond boundaries to integrate multiple perspectives, synthesise, and apply that knowledge to engage with subjects traditionally regarded as beyond the professions' remit; different organisations and agencies must work together to facilitate open cultures of supportive teamwork. End-users, whether clinical professionals or public contributors, should always be included. The benefits of these authentic collaborations will be evident in educational provisions, research and innovation, policy-making and clinical practice.

Conclusion: an all-hands-on-deck approach is needed

For safe and effective AI implementation in MIRT, there is a need to work collaboratively, be adequately trained and operationally prepared to assume new responsibilities and roles. New opportunities will be created in AI education, research, governance, and quality assurance. Recent research has highlighted the need for new professional roles, and clinical practitioners have already advocated for customised training, clearer guidance and leadership opportunities to enable them to engage in these new roles [74-77]. Different MIRT professionals may have different priorities and different needs regarding AI implementation in clinical practice [78]. However, they all

underline the need to work closely together with public contributors, academia, industry, professional bodies, to maximise the positive impact of AI and minimise its potential risks for patients and professionals. It is only through this collaboration that better clinical AI-enabled tools can be created for the benefit of everyone.

Reflexivity statement

Our team consists of radiologists ($n = 11$) of which one working in radiation oncology, radiographers ($n = 22$) of which 19 diagnostic and 3 therapeutic, computer scientists ($n = 2$), medical physicists ($n = 3$) one of which in radiotherapy and one in nuclear medicine departments, psychologists ($n = 2$), engineers ($n = 2$), one technical physician, one behavioural scientist, one implementation scientist, one cognitive science expert, one IT expert, and patients ($n = 2$). There are 27 men and 22 women of diverse ethnicity and age. Representatives from 5 professional bodies, 8 companies working in AI (of which 3 in large manufacturers and 5 in SMEs, 21 different Universities and 12 different countries around the world have participated in this work. Despite our different training, experiences, and current roles and professional backgrounds, we firmly believe in the power of collaboration and diverse representation in the MIRT AI ecosystem to create better clinical tools for the benefit of everyone.

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